

Savitribai Phule Pune University

M.Sc. Computer Science (2023)

CS-554-MJP : Lab Course on CS-551-MJ (Design and Analysis of Algorithms)

No. of Credits: 2	Teaching Scheme Theory: 4 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks
Prerequisite • Basic knowledge of algorithms and programming concepts • Data Structures and Advanced Data Structures • Basic Knowledge of Graphs and Algorithms • Basic knowledge of C/C++/ Java		
Objectives • To design the algorithms • To learn basic Algorithm Analysis techniques and understand the use of asymptotic notation • To Understand different design strategies • To Understand the use of data structures in improving algorithm performance • To critically analyze the efficiency of alternative algorithmic • To understand different algorithm design techniques. • To provide foundation in algorithm design and analysis • To develop the ability to understand and design algorithms in the context of space and time complexity.		
Course Outcomes On Completion of this course, student will be able to - CO1: Analyze worst-case running times of algorithms using asymptotic analysis. CO2: Compare between different data structures. Pick an appropriate data structure for a design situation. CO3: Ability to design algorithms using standard paradigms like: Greedy, Divide and Conquer, Dynamic Programming and Backtracking. CO4: Able to Explain the major graph algorithms and Employ graphs to model engineering problems, when appropriate. CO5: Able to Compare between different data structures and pick an appropriate data structure for a design situation.		
Assign No.	Name of Practical Assignment	
<u>1</u>	Write programs in C/C++/ Java to sort a list of n numbers in ascending order using selection sort, insertion sort, heap sort, radix sort. Determine the time required to sort and compare on basis of time complexity for different values of n.	
<u>2</u>	Write a program in C/C++/ Java to sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment	

	for different values of n, the number of elements in the list to be sorted. The elements can be read from a file or can be generated using the random number generator.
<u>3</u>	Write a program in C/C++/ Java to implement a Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted. The elements can be read from a file or can be generated using the random number generator.
<u>4</u>	Write a program in C/C++/ Java to implement Strassen's Matrix multiplication
<u>5</u>	Write a program in C/C++/ Java to find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm
<u>6</u>	Write a program in C/C++/ Java to find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm
<u>7</u>	Write a program in C/C++/ Java to from a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm
<u>8</u>	Write a program in C/C++/ Java to implement Knapsack problems using Greedy method
<u>9</u>	Write a program in C/C++/ Java to implement optimal binary search tree and also calculate the best case and worst case complexity.
<u>10</u>	Write a program in C/C++/ Java to implement huffman Code using greedy methods and also calculate the best case and worst case complexity.
<u>11</u>	Write a program in C/C++/ Java to find Minimum number of multiplications in Matrix Chain Multiplication
<u>12</u>	Write a Program in C/C++/Java to find only length of Longest Common Subsequence.
<u>13</u>	Write programs in C/C++/ Java to implement DFS and BFS. Compare the time complexity
<u>14</u>	Write a program in C/C++/ Java for finding Topological sorting for Directed Acyclic Graph (DAG)
<u>15</u>	Write a program in C/C++/ Java to determine if a given graph is a Hamiltonian cycle or not
<u>16</u>	Write a Java Program in C/C++/ Java to implement Traveling Salesman Problem using nearest neighbor algorithm
<u>17</u>	Write a program in C/C++/ Java a to implement Graph Coloring Algorithm
<u>18</u>	Write a program in C/C++/ Java to implement Sum of Subset by Backtracking
<u>19</u>	Write a program in C/C++/ Java to solve N Queens Problem using Backtracking
<u>20</u>	Write a program in C/C++/ Java to solve 4 Queens Problem using Backtracking
<u>21</u>	Write a program in C/C++/ Java to show board configuration of 4 queens problem
<u>22</u>	Write a program in C/C++/ Java to find out longest common subsequence from the given strings
<u>23</u>	Write a program in C/C++/ Java to find out live node, E node and dead node from a given graph
<u>24</u>	Write a program in C/C++/ Java to find out solution for travelling salesman problem using LCBB from a given matrix.
<u>25</u>	Write a program in C/C++/ Java to find out solution for 0/1 knapsack problem